WORKED SOLUTION

STEP I 1991, Question 11

Mechanics • Pulleys and Impulses

STEP I 1991, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s1-q11

The key idea

Each cycle has two stages. When the weight hits the pile it stops dead and the string goes slack, so the counterweight flies up and falls back freely under gravity. Then the string jerks taut and the two masses share momentum, and they move together with the constant acceleration from the first part until the weight lands again. Constant acceleration makes each stage symmetric in time. So every speed in one cycle is a fixed multiple of the speed in the cycle before, and a geometric progression follows straight away.

Solution

The constant acceleration

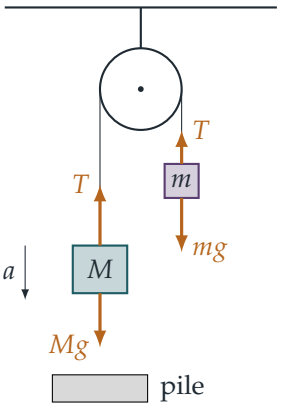
We model the weight and counterweight as particles, and the string as light and inextensible over a smooth light pulley. While the string is taut both particles move with the same speed v, the weight going down and the counterweight going up. Suppose that after moving a distance x from rest they have speed v. The weight has lost potential energy Mgx and the counterweight has gained mgx. The pulley is smooth and the string is light and inextensible, so no energy is lost and

$$\tfrac{1}{2}(M+m)v^2 = (M-m)gx.$$

Differentiating with respect to time, and using $\dot{x} = v$,

$$(M+m)v\dot{v} = (M-m)gv \quad \Longrightarrow \quad \dot{v} = g\,\frac{M-m}{M+m}$$

for $v \neq 0$. This is a constant. Call it a. The same result comes from Newton's second law: if T is the tension, then $Mg - T = Ma$ and $T - mg = ma$, and adding these gives $(M-m)g = (M+m)a$.



From one collision to the next

Just before a collision the weight is moving down with speed v, and the counterweight is moving up with the same speed v. Follow the motion in three stages.

The impact. The collision is completely inelastic and the pile is fixed, so the weight stops. The counterweight is still moving up at speed v, so the string goes slack. Tension then can't help the counterweight, and gravity alone pushes the weight onto the pile, so the weight stays put.

Free flight. With the string slack the counterweight is a projectile moving vertically. It rises and returns to the point where the string becomes taut again, which is where it was at the impact. Under constant acceleration g the motion is symmetric, so it arrives there moving *down* at speed v, after a time

$$\frac{2v}{g}.$$

The jerk. The string snaps taut with the counterweight moving down at v and the weight at rest. The masses must then move with a common speed u, the counterweight downwards and the weight upwards. By the given assumption, the momentum lost by the counterweight equals the momentum gained by the weight:

$$m(v-u) = Mu \quad \Longrightarrow \quad u = \frac{mv}{M+m}.$$

? **Ask yourself** Why can't we just use conservation of momentum for the whole system here?

The two masses move in opposite directions and the pulley exerts an impulse, so there is no single direction in which total momentum is conserved. What is true is that the impulsive tension has the same size J on both masses, since the string is light and the pulley smooth. So the counterweight loses momentum J along the string and the weight gains J along the string. That's exactly the statement we were allowed to assume.

Together again. Now the string is taut and the weight is moving up at speed u. By the first part, the acceleration has the constant magnitude a and is directed so that the heavier mass M goes down. So the weight slows to rest, then falls back to the pile. Constant acceleration makes this symmetric too: the weight reaches the pile at speed u, after a time

$$\frac{2u}{a}.$$

So just before the next collision the counterweight is moving with speed

$$u = \frac{mv}{M+m},$$

as required.

The times form a geometric progression

Write $k = \dfrac{m}{M+m}$, so $0 < k < 1$. Let v_0 be the speed just before the first impact. By the previous part the speed just before the $(n+1)$th impact is $v_n = k^n v_0$. Adding the two stages, the time from the $(n+1)$th impact to the next one is

$$t_n = \frac{2v_n}{g} + \frac{2kv_n}{a} = \left(\frac{2}{g} + \frac{2k}{a}\right) k^n v_0.$$

This is a constant times k^n, so the times between successive collisions form a geometric progression with common ratio $k = m/(M+m)$.

The total time

Let T_0 be the time from release to the first impact. The weight starts from rest with acceleration a, so $v_0 = aT_0$. Since $0 < k < 1$, the geometric series converges, and the total time is

$$T_0 + \sum_{n=0}^{\infty} t_n = T_0 + \left(\frac{2}{g} + \frac{2k}{a}\right) \frac{v_0}{1-k}.$$

Now $1 - k = \dfrac{M}{M+m}$, and $\dfrac{k}{a} = \dfrac{m}{M+m} \cdot \dfrac{M+m}{g(M-m)} = \dfrac{m}{g(M-m)}$. Hence

$$\frac{2}{g} + \frac{2k}{a} = \frac{2}{g}\left(1 + \frac{m}{M-m}\right) = \frac{2M}{g(M-m)},$$

$$\left(\frac{2}{g} + \frac{2k}{a}\right) \frac{v_0}{1-k} = \frac{2M}{g(M-m)} \cdot \frac{M+m}{M} v_0 = \frac{2v_0}{a} = 2T_0.$$

So there are infinitely many collisions, but they take only a finite total time. The weight finally comes to rest after a time

$$T_0 + 2T_0 = 3T_0,$$

which is three times the time from the start to the first impact.

Checks

- Take the limit $m \to 0$. Then $k \to 0$ and $a \to g$, so the weight never lifts again after the first impact. The only remaining time is the counterweight's flight, $2v_0/g = 2gT_0/g = 2T_0$, and the total is $3T_0$, as the general formula predicts.
- We stepped the motion numerically for random M, m and drop heights (for example $M = 2.403$, $m = 1.571$, so $k = 0.3953$). The speed ratios and gap ratios both came out as 0.3953, and the total time, including the geometric tail, came out as $2.99998\,T_0$.

Common slips

- Forgetting the free-flight stage. The counterweight carries on upwards after the impact, and the cycle includes the time $2v/g$ it spends as a projectile.
- Using a instead of g for the slack phase. Once the string is slack the counterweight feels only gravity.
- Writing "total momentum is conserved" at the jerk with both velocities in the same direction. The pulley turns the string, so it is the momentum *along the string* that is shared.

Beyond STEP

Infinitely many events in a finite time, like this piledriver or a bouncing ball, are called Zeno behaviour. They turn up in the theory of hybrid systems, where smooth motion is interrupted by sudden jumps. The jerk in the string is a simple impulsive constraint. Lagrangian mechanics treats such constraints systematically in the first undergraduate year. For more on impulses, collisions and constrained motion, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

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